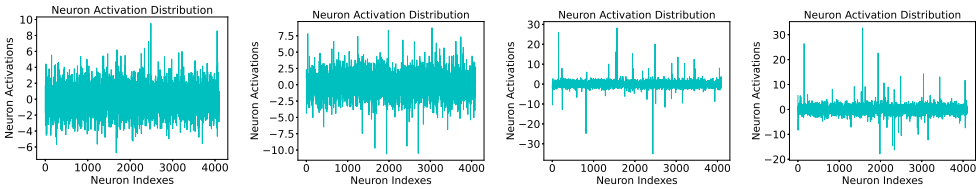

Figure 6: Activation distributions of four selected tokens in LLaMA-7B.

access the hidden states. Additionally, our proposal is a sampling-based approach, necessitating the generation of multiple outputs, which introduces additional inference overhead for accurate hallucination detection. Furthermore, another limitation is that we only focus on hallucination detection in this study and have not yet mitigate the hallucination with our proposal. In the future work, we hope to reduce the inference cost of our proposal and leverage the EigenScore to mitigate hallucinations. We believe the proposed EigenScore is a strong baseline for detecting hallucination and we invite further researchers to utilize and enhance our proposal.

H Cases Study

H.1 Hallucination Detection Cases

We show several cases with LLaMA-7B and NQ dataset. The number of generation is set to $N = 10$. We determine the optimal detection threshold by maximizing the **G-Mean** value, which is defined as **G-Mean** $= \sqrt{TPR * (1 - FPR)}$. The hallucination detection thresholds for different metrics are Perplexity: 0.535; LN-Entropy: 0.153; LexicalSimilarity 0.489; SelfCheckGPT: 0.168; EigenScore: -1.74. For LexicalSimilarity, a score larger than the threshold indicates non-hallucination. For other metrics, a score smaller than the thresholds indicates non-hallucination. Green indicates non-hallucination answer and Red indicates hallucination answer generated by LLM. ✓ (✗) indicates that the hallucination is (not) correctly identified by the metric.

Question: the german princes who chose the holy roman empire were called
GTAns: prince-electors
LLMAns: electors
BatchGenerations: ['electors', 'electors', 'electors', 'electors', 'electors', 'electors', 'electors', 'electors', 'electors', 'electors']
Perplexity: 0.361 ✓
LN-Entropy: 0.027 ✓
LexicalSimilarity: 1.0 ✓
SentBERTScore: 0.0 ✓
EigenScore: -2.63 ✓
EigenValue: [4.87719579e+00 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03]

Question: where is fe best absorbed in the body
GTAns: in the duodenum
LLMAns: in the small intestine
BatchGenerations: ['in the liver', 'small intestine', 'in the intestines', 'the small intestine', 'the small intestine', 'in the liver', 'small intestine', 'fat', 'in the small intestine', 'fatty tissues']
Perplexity: 0.641 ✓
LN-Entropy: 0.213 ✓
LexicalSimilarity: 0.357 ✓
SentBERTScore: 0.258 ✓
EigenScore: -1.40 ✓
EigenValue: [3.71561676e+00 4.34496729e-01 3.77751922e-01 1.75326593e-01 9.92596975e-02 4.20723353e-02 2.49385766e-02 1.00000000e-03 1.00000000e-03 1.00000000e-03]

Question: who did the united states win its independence from
GTAns: the British Empire
LLMAns: britain
BatchGenerations: ['britain', 'england', 'great britain', 'great britain', 'england', 'england', 'england', 'england', 'great britain', 'great britain']
Perplexity: 0.598 ✗
LN-Entropy: 0.266 ✗
LexicalSimilarity: 0.415 ✗
SentBERTScore: 0.397 ✗
EigenScore: -2.23 ✓
EigenValue: [4.46843402e+00 2.82423429e-01 3.88702191e-02 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03]

Question: who won the most stanley cups in history
GTAns: Montreal Canadiens
LLMAns: the detroit red wings
BatchGenerations: ['the detroit red wings', 'the detroit red wings', 'the detroit red wings', 'the detroit red wings', 'the detroit red wings', 'the detroit red wings', 'the detroit red wings', 'the detroit red wings', 'the detroit red wings', 'the detroit red wings']
Perplexity: 0.366 ✗
LN-Entropy: 0.025 ✗
LexicalSimilarity: 1.0 ✗
SentBERTScore: 0.0 ✗
EigenScore: -2.63 ✗
EigenValue: [5.23534401e+00 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03]

Question: what is the second book in the alchemyst series
GTAns: The Magician
LLMAns: the alchemyst: the secret of the immortal Nicholas flamel
BatchGenerations: ['the magician in the middle', "the magician's nephew", 'the magician', 'the alchemyst', 'the magician', 'the alchemyst', 'the magician in the middle', 'the magician in amsterdam', 'the magician in the middle', 'the secret of the near witch']
Perplexity: 0.459 ✗
LN-Entropy: 0.207 ✓
LexicalSimilarity: 0.499 ✗
SentBERTScore: 0.431 ✓
EigenScore: -1.41 ✓
EigenValue: [2.09138499e+00 6.95605781e-01 3.85931973e-01 3.40671669e-01 2.15372994e-01 1.77304781e-01 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03]

Question: a good that can be used in place of another good
GTAns: A substitute good
LLMAns: substitute
BatchGenerations: ['a substitute', 'a substitute', 'substitute', 'a substitute', 'substitute', 'substitute', 'substitute', 'substitute', 'substitute', 'substitute']
Perplexity: 0.614 ✗
LN-Entropy: 0.194 ✗
LexicalSimilarity: 0.844 ✓
SentBERTScore: 0.076 ✓
EigenScore: -2.40 ✓
EigenValue: [5.11365925e+00 1.75884104e-01 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03]

Question: who said i will not go quietly into the night
GTAns: Welsh poet Dylan Thomas
LLMAns: douglas macarthur
BatchGenerations: ['douglas macarthur', 'elvis presley', 'douglas macarthur', 'douglas macarthur', 'douglas macarthur', 'robin williams', 'robert f. kennedy', 'robin williams', 'robert f kennedy', 'Dwight Eisenhower']
Perplexity: 0.502 ✗
LN-Entropy: 0.121 ✗
LexicalSimilarity: 0.178 ✓
SentBERTScore: 0.337 ✓
EigenScore: -1.61 ✓
EigenValue: [3.32824135e+00 5.87944819e-01 3.70390066e-01 1.70849836e-01 1.17707239e-01 5.17925563e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03]

Question: when did they stop saying mass in latin
GTAns: from 1967 onwards
LLMAns: 1969
BatchGenerations: ['in 1969', '1969', 'around 1969', 'in 1965', 'in 1969', '1969', 'In 1969, the Roman Missal was changed from Latin to the vernacular.', '1969', '1969 when the second vatican council was held', '1969']
Perplexity: 0.631 ✓
LN-Entropy: 0.152 ✗
LexicalSimilarity: 0.540 ✗
SentBERTScore: 0.273 ✓
EigenScore: -1.59 ✓
EigenValue: [3.80408192e+00 4.83987672e-01 3.03207580e-01 8.80366008e-02 6.59790286e-02 3.26742841e-02 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03]

Question: where does the highway of heroes start and end
GTAns: Glen Miller Road in Trenton
LLMAns: it starts in ottawa and ends in toronto
BatchGenerations: ['start in ottawa and end in toronto', 'it starts in toronto and ends in ottawa', "it starts at the queen's park gate at queen's park and ends at toronto's coronation park", "it starts in toronto and ends at queen's park", 'starts in Toronto and ends in Trenton', 'it starts at the CNE and ends at the cenotaph in downtown toronto', 'it starts in downtown toronto and ends in ottawa', 'it starts in the city of Toronto and ends in the city of Hamilton', 'it starts in toronto and ends in ottawa', 'it starts in Toronto and ends in Ottawa']
Perplexity: 0.657 ✓
LN-Entropy: 0.260 ✓
LexicalSimilarity: 0.594 ✗
SentBERTScore: 0.149 ✗
EigenScore: -1.05 ✓
EigenValue: [3.31983018e+00 3.98560810e-01 2.17094299e-01 2.06965709e-01 1.53575354e-01 1.27925588e-01 7.82365136e-02 3.28158137e-02 1.01995086e-02 1.00000000e-03]

Question: which body part(s) occupy the greatest portion of the primary motor cortex
GTAns: face
LLMAns: the hand
BatchGenerations: ['the hand', 'the hand', 'the fingers', 'the hand', 'the hand', 'the hand and the foot', 'the hand', 'the hand and the foot', 'the hand and face', 'arms and legs']
Perplexity: 1.41 ✓
LN-Entropy: 0.300 ✓
LexicalSimilarity: 0.568 ✗
SentBERTScore: 0.163 ✗
EigenScore: -1.69 ✓
EigenValue: [3.76273036e+00 6.16284067e-01 1.96541049e-01 1.73505005e-01 1.28407153e-01 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03 1.00000000e-03]